

Luis Angel Guerrero Hoyos

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Professional Summary

GIS Analyst with 6+ years of experience delivering engineering-quality mapping, geospatial analysis, and geoprocessing automation for infrastructure, energy, and natural-resources projects. Skilled in QGIS and ArcGIS Pro cartographic production, Python geoprocessing automation (GDAL, GeoPandas, Rasterio), and LiDAR/terrain analysis (point-cloud registration, contour and elevation-product generation). Experienced processing UAV and satellite-derived imagery and maintaining project geodatabases across multidisciplinary engineering teams. Holds an MSc in Civil Engineering (University of Washington) and a BSc in Geological Engineering.

Skills

- **GIS Software:** QGIS (Processing Models, styling libraries, map templates), ArcGIS Pro (arcpy, layouts, map series), ArcGIS Online
- **Geoprocessing Automation:** Python (GDAL, GeoPandas, Rasterio, Shapely), raster/vector QC (clipping, mosaicking, reprojection, topology validation, buffering, attribute validation)
- **Terrain & LiDAR:** Point-cloud processing/registration (PDAL, CloudCompare/Cloudcompy), classified surfaces, contour/elevation products, UAV LiDAR evaluation
- **Remote Sensing & Data:** UAV/orthophotography workflows, Sentinel-1 SAR/InSAR (ISCE+, Dolphin, PyGMTSAR), geodatabases, GeoJSON, GeoPackage, Shapefile, DEMs, CRS, SQL
- **Project Management Skills:** Cross-disciplinary collaboration with civil, structural, and environmental teams; SOP documentation and data management

Experience

Staff Engineer – GIS & Remote Sensing (July 2025 – Present)

Shannon and Wilson Inc. 

- Lead developer of SWLC3D, the firm's LiDAR point-cloud change-detection pipeline (ICP registration and M3C2 differencing) using PDAL, CloudCompare/Cloudcompy, and arcpy, producing shapefile deliverables for ArcGIS Pro.
- Develop cartographic products, standardized map templates, and layer/geodatabase management across multi-site programs, including 97 ArcGIS Pro layers across 10 sites for a pipeline monitoring program.
- Built cross-section map-series export and raster-sampling automation (Python) for a hydropower dam-monitoring program.

Research Assistant & Geospatial Scientist (August 2023 – June 2025)

University of Washington 

- Built Python geoprocessing workflows (GeoPandas, Rasterio, GDAL) integrating LiDAR elevation data, Sentinel-1 imagery, and vector datasets to study earthquake-induced landslides at regional scale; completed InSAR Processing (ISCE+) training, EarthScope Consortium & NSF (2024).

GIS & InSAR Monitoring Analyst (July 2021 – August 2023)

Hexagon Geospatial 

- Delivered real-time geospatial terrain monitoring of open-pit mine slopes and tailings dams, integrating GNSS field data collection and ground-based radar into centralized geodatabases across mining operations in Latin and North America.

Geological Engineer – GIS & Resource Data Management (January 2019 – July 2021)

South32 

- Managed GIS geodatabases and drill-hole spatial data for mineral resource estimation; coordinated field data collection and produced mapping deliverables supporting engineering decision-making.

Education

- Master of Science in Civil Engineering
University of Washington
June 2025

- Bachelor of Science in Geological Engineering
Universidad Nacional de Colombia (UNALMED)
May 2019

Awards & Certifications

- InSAR Processing and Analysis (ISCE+) – EarthScope Consortium & NSF, August 2024
- CEE Departmental Graduate Fellowship – University of Washington, September 2023
- InSAR and RAR Radar Monitoring – University of Arizona, December 2020

Languages

- English
- Spanish (Native)